EXPERIMENT 9: Schmitt Trigger (Comparator with Hysteresis)

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Objective

To study the operation of a Schmitt trigger (comparator with positive feedback), measure its switching thresholds and hysteresis at two input bias levels (0 V and 2.5 V), and compare experimental waveforms with LTspice simulation results.

Introduction and Theory

A Schmitt trigger is a comparator with positive feedback that implements hysteresis: the switching threshold for the rising input is different from the threshold for the falling input. This prevents chatter when the input varies around the switching point.

Consider a non-inverting Schmitt trigger built from an op-amp with resistors R_1 (feedback from output to the non-inverting input) and R_2 (from non-inverting input to ground). Let the output saturate to V_{OH} (high) and V_{OL} (low). The switching thresholds are:

$$V_{TH}^{\uparrow} = V_{UP} = V_{OH} \cdot \frac{R_2}{R_1 + R_2},$$
$$V_{TH}^{\downarrow} = V_{DOWN} = V_{OL} \cdot \frac{R_2}{R_1 + R_2}.$$

The hysteresis width (difference between upper and lower thresholds) is

$$\Delta V = V_{UP} - V_{DOWN} = (V_{OH} - V_{OL}) \frac{R_2}{R_1 + R_2}.$$

For an inverting Schmitt trigger (input at the inverting input while the non-inverting input is tied to a reference), the thresholds are determined similarly by the feedback divider and the reference level; the core idea is the same: thresholds depend on output saturation levels and the resistor ratio.

Apparatus and Circuit Details

- Op-amp/Comparator: 741 (or generic op-amp used in the experiment)
- Resistors: R_1 and R_2 (values used in experiment should be listed here)
- Power supply: single supply (e.g., 0–5V) or dual supply (as used); record V_{CC} used in the experiment.

Procedure

1. Assemble the Schmitt trigger circuit on the breadboard as shown in the lab manual (see attached PDF reference).
2. Apply a slow triangular wave (or slowly varying DC ramp) to the input and observe the input and output on the oscilloscope.
3. Record the input voltage at which the output switches from low to high (V_{UP}) and the voltage at which it switches from high to low (V_{DOWN}) for the two bias conditions (0 V and 2.5 V) used in the experiment.
4. Capture photographs of the experimental setup and the oscilloscope traces.
5. Run LTspice simulation of the same circuit, export waveform images and compare.

Reference / Datasheet

See lab reference: *Schmitt trigger manual* (included in workspace): Scmitt trigger manual (1).pdf

Experimental Results: 0 V bias

Below are the photographs and oscilloscope captures for the experiment performed with the reference/input bias at 0 V.

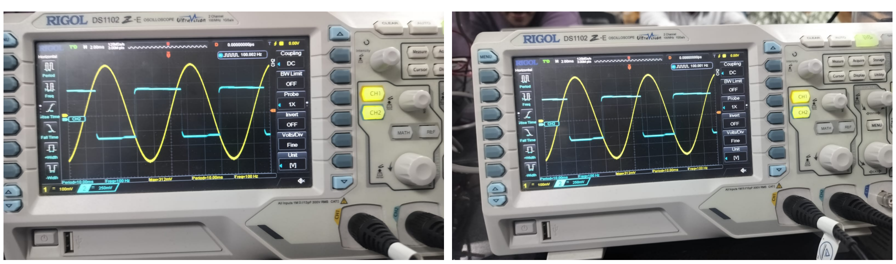

Figure 1: Setup and oscilloscope traces (0 V bias) — sample images

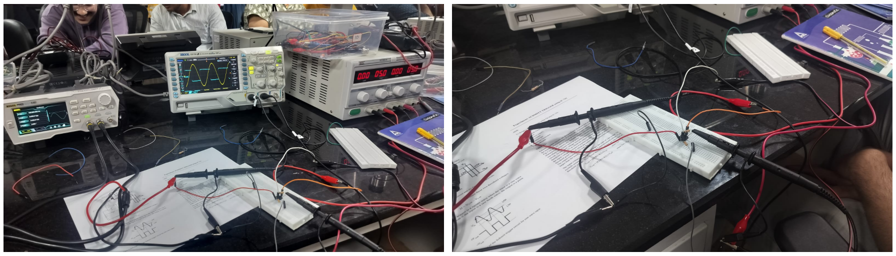

Figure 2: More traces / measurement photos (0 V)

Experimental Results: 2.5 V bias

Measurements taken with the input/bias set to 2.5 V.

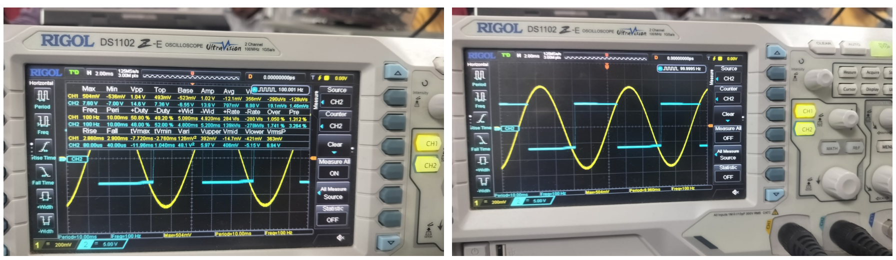

Figure 3: Setup and oscilloscope traces (2.5 V bias) — sample images

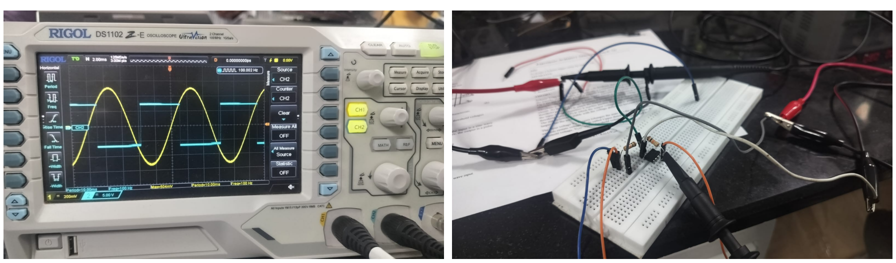

Figure 4: More traces / measurement photos (2.5 V)

LTspice Simulation Results

The LTspice exported screenshots for the Schmitt trigger simulation are included below. Compare the simulated switching thresholds and hysteresis with experimental values.

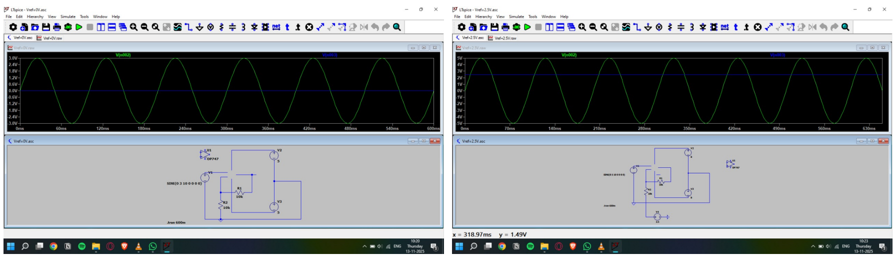

Figure 5: LTspice waveform exports for the Schmitt trigger simulation

Analysis and Conclusion

From the oscilloscope captures determine V_{UP} and V_{DOWN} for both bias conditions (0 V and 2.5 V). Using measured output saturation levels V_{OH} and V_{OL} and known resistor values R_1, R_2, compare the theoretical thresholds:

$$V_{UP}^{theory} = V_{OH} fracR_2R_1 + R_2, qquadV_{DOWN}^{theory} = V_{OL} fracR_2R_1 + R_2.$$

Compute the experimental hysteresis width $DeltaV_{exp} = V_{UP}^{exp} - V_{DOWN}^{exp}$ and compare with $DeltaV_{theory}$.